Comparative R&D Investigation & Cooking-Performance Optimization for Imported Tanzanian Toor Dal (Cajanus cajan)

Phase 1 Diagnostic Research Scope of Work, Commercial Terms & Technical Framework | Ref: FNQ-PR-2026-001 (Rev 8.0)

Client / Sponsor:	**Commercial Pulse Processing Facility Management & Technical Team (India)**
Lead Chemical & Process Engineer:	Sourish Senapati, FinnoaQ Agro-Industrial Technologies Ltd. (Kolkata, WB)
Academic Guidance & Scientific Advisory:	Dr. Archisman Ray, Senior Research Fellow, Dept. of Chemical Engineering, IIT Kharagpur
Project Date & Execution Baseline:	16 October 2026 | Execution Mode: Offline-First Native Engineering | Revision 8.0

1. Executive Summary & Commercial Rationale

Imported Tanzanian split, dehulled pigeon pea (Cajanus cajan)—commonly traded as Toor Dal—presents a compelling raw material procurement price advantage of approximately ₹18–22/kg relative to domestic Indian benchmark varieties. However, commercial uptake among Indian industrial millers and retail consumers is severely constrained by an acute cooking latency defect: under standard household pressure-cooking conditions, Tanzanian toor dal requires approximately 6 whistles (~45–55 minutes total cooking cycle) compared to 3 whistles (~20–25 minutes) for comparable domestic Indian toor dal. Even after extended thermal boiling, the imported grains frequently retain a persistent hard core, exhibit non-uniform softening, and display poor culinary mouthfeel.

This proposal establishes an authoritative, five-stage Phase 1 Diagnostic Research Program designed to scientifically determine the root biophysical, structural, and chemical mechanisms responsible for this cooking-time disparity. Operating under strict engineering governance, this study will generate empirical laboratory evidence across proximate composition, cell-wall pectin fractionation, mineral cross-linking, starch gelatinization, and hydration kinetics to provide a defensible foundation for subsequent commercial conditioning and industrial processing.

2. Phase 1 Technical Investigation Scope of Work

The Phase 1 investigation replaces qualitative household observations with rigorous, objective, and reproducible analytical methodologies adhering to AOAC, ISO, BIS, and FSSAI standards. Both the imported Tanzanian test lot and a representative domestic Indian benchmark lot will be evaluated simultaneously across 11 synchronized analytical work packages:

Module Ref	Work Package Description	Analytical Methodology & Instrumentation	Indicative Fee (INR)
WP-01	Sample Preparation & Physical Characterization	Grain dimensions (L/W ratio), 1000-grain weight, bulk density, broken %, defect count, raw hardness via digital micrometer and sieving.	₹20,000
WP-02	Proximate Chemical Composition	Standard AOAC / FSSAI protocols: moisture (vacuum oven), crude protein (Kjeldahl), crude fat (Soxhlet), crude fiber, total ash, and available carbohydrate.	₹7,000
WP-03	Starch & Amylose Profiling	Total starch determination, amylose-to-amylopectin ratio via iodine spectrophotometry, and enzymatic starch availability assays.	₹5,000
WP-04	DSC Gelatinization & Thermal Behavior	Differential Scanning Calorimetry (DSC): onset (To), peak (Tp), concluding (Tc) gelatinization temperatures, transition enthalpy (ΔH), and pasting behavior.	₹20,000

Module Ref	Work Package Description	Analytical Methodology & Instrumentation	Indicative Fee (INR)
WP-05	Hydration Kinetics & Water Absorption Index	Multi-point isothermal soaking curves (0, 15, 30, 60, 120, 240, 480 min) at 25°C, 45°C, and 65°C; gravimetric Water Absorption Index (WAI) and volume swelling.	₹10,000
WP-06	Pectin & Cell-Wall Fractionation	Sequential chemical fractionation: water-soluble pectin, chelator-soluble pectin (EDTA), and alkali-soluble pectin; degree of esterification; cell-wall rigidity analysis.	₹18,000
WP-07	Mineral Profile & Phytate Interactions	ICP-OES / AAS determination of divalent cations (Ca^{2+}, Mg^{2+}), iron, phosphorus, and spectrophotometric phytic acid quantification; phytate:cation molar ratios.	₹16,000
WP-08	Instrumental Hardness & Texture Profile	Texture Profile Analysis (TPA) on a calibrated Lloyd/Stable Micro Systems Texture Analyzer: raw breaking force, soaked firmness, cooked peak penetration resistance.	₹17,000
WP-09	Standardized Cooking-Performance Trials	Instrumented laboratory pressure vessel trials with calibrated temperature/pressure logging; determination of objective cooking endpoint (cooked core % < 2.0%).	₹20,000
WP-10	Integrated Data Analysis & Root-Cause Matrix	Multi-factor statistical correlation and analysis of variance (ANOVA) linking physicochemical parameters to cooking latency; candidate ranking.	₹20,000
WP-11	Comprehensive Technical R&D Report	Publication-grade final dossier synthesizing all raw data, statistical matrices, biophysical mechanisms, and formal Phase 2 processing recommendations.	₹20,000

3. Scientific Evidence Hierarchy & Interpretation Principles

To ensure scientific and commercial defensibility, FinnoaQ enforces a strict four-tier evidence hierarchy. No causal mechanism will be reported as established fact unless it is experimentally demonstrated through replicated wet-lab testing:

Evidence Level	Operational Definition	Project Governance Rule
Level 1: Measured Result	Direct empirical physical or chemical measurement obtained from verified laboratory instruments under standard operating procedures.	Reported with raw data, standard deviation, and analytical method citation.
Level 2: Observed Association	Statistically significant correlation observed between a measured parameter (e.g., pectin-associated Ca^{2+}) and cooking time.	Categorized as 'Strongly Associated Factor'; cannot assert standalone causality.
Level 3: Scientific Hypothesis	Biophysical or chemical mechanism proposed to explain observed associations (e.g., middle lamella pectate cross-linking).	Explicitly labeled 'HYPOTHESIS'; carried forward for Phase 2 validation.
Level 4: Phase 2 Validation	Controlled intervention trial demonstrating that targeted treatment reverses the defect under industrial conditions.	Required before commercial process guarantees or scale-up commitments.

Important Scope Boundary: Phase 1 is strictly a diagnostic R&D investigation. It does not provide commercial product guarantees, commercial processing parameters, or final construction engineering. These activities are addressed in subsequent phased offerings.

4. Candidate Interventions for Follow-On Phase 2 Evaluation

Based on Phase 1 diagnostic findings, Phase 2 will experimentally screen targeted interventions to accelerate cooking times. The relevance of each candidate will be determined strictly by the empirical Phase 1 evidence:

Intervention Category	Target Mechanism & Physiological Action	Regulatory & Processing Considerations
Food-Grade Mineral Conditioning	Alkaline salt soaking (Sodium Bicarbonate, Sodium Citrate) to chelate bound Ca^{2+}/Mg^{2+} from insoluble pectates in the middle lamella, promoting cotyledon cell separation.	Strict compliance with FSSAI Food Safety and Standards Regulations (Table 3 GMP limits); leaves zero chemical off-flavor or residue.
Controlled Hydrothermal Tempering	Precision temperature-programmed soaking (45°C–65°C) to accelerate moisture diffusion without inducing premature starch gelatinization or cotyledon cracking.	Requires continuous closed-loop water temperature control; minimal solids leaching (<1.5% w/w).
Targeted Thermal / Microwave Pre-Treatment	Rapid dry dielectric or infrared heating to create micro-fissures in dense cotyledon cell layers, facilitating rapid capillary water ingress.	Requires precise exposure limits to prevent protein denaturation, darkening, or seed coat discoloration.
Enzymatic Cell-Wall Attenuation	Pectinase / cellulase micro-dosing to selectively hydrolyze structural cell-wall polysaccharides and enhance cellular permeability.	Enzyme inactivation requirements; economics and processing costs evaluated in techno-economic study.

5. Commercial Terms, Turnaround Schedule & Milestones

FinnoaQ operates on an auditable milestone-based commercial structure. The total professional fee for the complete Phase 1 Diagnostic R&D Program is ₹1,73,000 (Rupees One Lakh Seventy-Three Thousand Only) plus applicable GST:

Project Milestone	Scope Description & Deliverable	Payment %	Fee Amount (INR)
Milestone 1: Project Initiation	Formal execution of onboarding agreement; assignment of unique lot custody tracking numbers; physical sample receipt and preparation.	40%	₹69,200
Milestone 2: Laboratory Completion	Completion of all physical, proximate, starch, thermal (DSC), hydration, pectin, mineral (ICP-OES), texture, and standardized cooking trials; interim data transmittal.	40%	₹69,200
Milestone 3: Final Report Release	Delivery and formal presentation of Comprehensive Technical R&D Report, Statistical Root-Cause Correlation Matrix, and Prioritized Phase 2 Technology Recommendations.	20%	₹34,600

Key Commercial Conditions:

• Commercial Working Ceiling: For procurement planning, an expenditure ceiling of up to ₹2,00,000 + applicable GST is established. Any additional specialized testing requires prior written client authorization.

• Taxes: Goods and Services Tax (GST) will be charged additionally as applicable under prevailing statutory rates.

• Turnaround Time: 20 to 25 business days from verified receipt of representative samples at our Kolkata analytical reception.

• Fast-Track Interim Summary: An executive Stage 1 preliminary data brief will be delivered within 7 business days of sample arrival.

• Offer Validity: This commercial proposal is valid for 30 calendar days from the date of issuance.

6. Sample Dispatch & Chain of Custody Protocol

To ensure experimental validity and statistical rigor, the client will supply representative sample lots adhering to the following protocol:

1. Required Quantity: 2.5 kg to 3.0 kg of the target imported Tanzanian commercial toor dal lot, and 2.5 kg to 3.0 kg of standard domestic Indian benchmark dal.

2. Packaging: Samples must be double-bagged in food-grade, moisture-barrier polyethylene bags and sealed in rigid outer cartons.
3. Lot Documentation: Each consignment must include batch/lot ID, port of entry, date of import/milling, and known storage conditions.
4. Consignment Address: Central Laboratory Reception, FinnoaQ Agro-Industrial Technologies Ltd., Kolkata, West Bengal (Attn: Sourish Senapati).
5. Retention: A 500 g sealed reference sample from each lot will be securely archived at 20°C / 55% RH for 12 months.

7. Industrial Scale-Up Roadmap: Phased Integration

To prevent premature capital expenditure, FinnoaQ structures industrial pulse processing into distinct, de-risked decision gates:

• Phase 1 (Current Base Scope): Diagnostic baseline & biophysical root-cause identification (₹1.73 Lakh + GST, 20–25 days).

• Optional Phase 2A (Laboratory Validation): Controlled formulation screening across candidate conditioning agents; bench-scale cooking confirmation; sensory, nutritional, and shelf-life assessment.

• Optional Phase 2B (Commercial Plant FEED): Conceptual Front-End Engineering Design for a continuous 15,000 kg/day (15 TPD) industrial dal conditioning plant. FinnoaQ maintains an integrated 3D SolidWorks parametric model (PLANT-15T, 12 interconnected systems, 27.08 tonnes B-rep assembly), continuous fluid-bed dryer psychrometric models, and converged Aspen Plus V14 Zero-Liquid-Discharge (ZLD) falling-film evaporator balances (steam economy 2.38). This engineering baseline will be tailored to the client's verified site layout upon successful completion of Phase 1.

8. Master Engineering Drawing Annexures (15 TPD Industrial Plant Concept)

To substantiate the industrial viability of the proposed continuous hydration and drying technology, FinnoaQ maintains a fully developed Front-End Engineering Design (FEED) package. The following authentic engineering drawings, generated via Autodesk AutoCAD 2025 (accoreconsole native vector engine) and SolidWorks 2025 SP01.2 parametric solid modeling, are attached as official project annexures:

Annexure Ref	Drawing Number & Title	Design Standard & Scope Description	Format & Native Tool Provenance
Annexure A	DWG-FIN-15T-001 Process Flow Diagram (PFD)	12 reconciled streams, material balance flags, equipment silhouettes, CIP & ZLD connections.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)
Annexure B	DWG-FIN-15T-002 Heat & Material Balance Schedule	Full A2 stream table (solids, water, enthalpy), exact 0.000% dry solids closure, utility duties.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)
Annexure C	DWG-FIN-15T-003 P&ID: Soaking Battery & Dosing Skid	ISA-5.1 dual 7.5m3 soak tanks TK-101A/B, dosing skid SK-100, LIC/TIC/AIC/ZIC control loops.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)
Annexure D	DWG-FIN-15T-004 Site Master Plot Plan & Hygienic Zoning	75m x 45m plot, 52m x 24m building, 4 ISO 22000 hygiene zones, room HVAC & area schedule.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)

Annexure Ref	Drawing Number & Title	Design Standard & Scope Description	Format & Native Tool Provenance
Annexure E	DWG-FIN-15T-007 Equipment GA: Soak Vessel TK-101A/B	ASME Section VIII Div 1, 2000mm ID, 60° cone, 4.0 kW hydrofoil agitator, nozzle schedule N1-N6/M1.	AutoCAD 2025 + SolidWorks ISO A2 (monochrome.ctb)
Annexure F	DWG-FIN-15T-008 Equipment GA: Fluid Bed Dryer DR-101	14m x 1m SS304 deck, 3 psychrometric zones, dual NFPA 68 burst panels, 8 vibration springs.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)
Annexure G	DWG-FIN-15T-010 P&ID: Triple-Effect ZLD Evaporator	25 KLD falling-film calandrias, vacuum condenser, distillate recovery loop, steam economy 2.38.	AutoCAD 2025 + Aspen Plus ISO A2 (monochrome.ctb)
Annexure H	DWG-FIN-15T-011 Electrical Single Line Diagram & SCADA	11kV / 415V 500kVA TR-01, 250kVA DG, 11 MCC feeders, ISA-95 SCADA Profinet ring topology.	AutoCAD 2025 (Vector PDF) ISO A2 (monochrome.ctb)
Annexure I	DWG-FIN-15T-014 Plant GA Master Assembly Drawing	SolidWorks 3D B-Rep assembly (PLANT-15T, 12 systems, 27.08 tonnes), isometric & orthographic GAs.	SolidWorks 2025 SP01.2 A1 Master Assembly Drawing
Master Book	FINNOAQ_15TPD_CONCEPTUAL_FEED_DRAWINGS_REV_V8.pdf	Consolidated 8-sheet conceptual FEED package compiled into a single publication-grade book.	AutoCAD 2025 Native Plot 625,497 Bytes (Verified)
Master Dossier (50 Pages)	FINNOAQ_15TPD_MASTER_INTEGRATED_ENGINEERING_DOCUMENT.pdf	Complete 50-page Master Integrated Engineering Dossier: AutoCAD 2D Drawings paired with SolidWorks 3D Models in Upright Human Orientation, 8 Dedicated Joined Plant Exhibition Sheets, Dual-Pane Suite, and Aspen V14 simulation.	AutoCAD + SolidWorks + Aspen 29.67 MB | SHA-256 Verified

Engineering Provenance Note: All attached drawings strictly observe engineering integrity under FinnoaQ Governance Law. Drawings originate from native AutoCAD accoreconsole plotting and SolidWorks B-Rep models with 0% synthetic approximations. All plant engineering deliverables remain conceptual FEED baselines under Gate G-A hold, pending empirical Phase 1 validation.

9. Project Acceptance & Start Authorization Form

To confirm acceptance of this proposal and authorize commencement of Phase 1, please complete and sign below, or issue a formal Purchase Order / Work Order referencing Document Ref: FNQ-PR-2026-001 (Rev 8.0):

Client Corporate Entity Name:	
Authorized Representative Name & Title:	

Client Corporate Entity Name:	
Corporate Registered Address & GSTIN:	
Authorized Signature & Corporate Seal:	_____
Execution Date:	
Target Sample Dispatch Date:	